UniGuide

Quality Assurance Testing Document (Refined)

UMHackathon 2026, Final Round, Domain 1

Team: Breaking Bank, Team 108
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Table of Contents

1. Executive Summary 2
2. Test Strategy 2
3. Test Plan & Coverage Matrix 3
4. Test Execution Evidence 6
5. Defect Log 7
6. Risk Register 8
7. Architecture Trade-Offs 9
8. Resilience Test Cases (Deep Dive) 10
9. Quality Gates 10
10. Future Test Plan (Post-Finals) 11
11. Terms & Definitions 11
12. Conclusion 13

1. Executive Summary

This document is the refined Quality Assurance and Testing Document for UniGuide at UMHackathon 2026 Finals. It updates the preliminary QATD (submitted 26 April) with the additional features shipped during the online and physical sub-phases of the final round, and replaces the prelim defect list with a current one.

Quality confidence at finals submission rests on four pillars: (a) a fully-typed schema layer that catches contract drift at build and request time (TypeScript strict plus Zod runtime validation); (b) one resilience smoke test that exercises the AI mock-fallback paths; (c) extensive manual demo runs verifying every PRD feature end-to-end on the live deploy; and (d) deliberate engineering of resilience layers so that any single AI failure degrades gracefully rather than orphaning an application.

The team explicitly acknowledges that automated test coverage at finals submission is light, namely one integration smoke test. This is by design under the time-box. The post-finals quality plan ramps integration and end-to-end test coverage to a full pyramid by Q3 2026. A glossary of recurring terms is provided at the end (Section 11).

2. Test Strategy

Three layers, each calibrated to the hackathon's risk profile.

- **Static layer (always on):** TypeScript strict mode, Zod runtime validation on every API request body, prompt-hash recording on every AI call. Catches type drift, malformed requests, schema regressions, and prompt-version changes.

- **Resilience-fixture layer (automated):** tests/glm-mock.test.ts confirms that with no AI keys configured, every GLM and ILMU call site returns the recorded fixture rather than throwing. This is the single most important automated check for the demo: no judge should ever see a 5xx mid-pitch.
- **Manual happy-path plus edge-case sweeps (per release):** Before each significant push, the team runs a scripted sequence covering the full student-to-coordinator-to-letter loop, plus the new features added in this build (parseDocument extraction, SOP-grounded briefing, plan-mode preview-step, coordinator coassist, student chat plus escalation). Results are captured in Section 4 below.

3. Test Plan & Coverage Matrix

All PRD features (F1 to F28) plus the six features added during the finals build (F29 to F34) traced to test cases and their current status. Green rows are PASS in the most recent test run.

ID	Feature	Test case	Status
F1	Three role surfaces (student, coordinator, admin)	Manual sweep across all three surfaces; demo tile sign-in resets data each time	PASS, all three roles render their landing page
F2	AI step generation turn-by-turn via GLM nextStep	Start a Scholarship application, walk Steps 1 to 6, confirm next-step is grounded in SOP citations	PASS, citations visible on every step in the live deploy
F3	SOP citation chips clickable on every AI step	Click any citation; SOP viewer opens scrolled to that section	PASS
F4	Coordinator briefing with extracted facts, flags, recommendation, confidence	Submit a borderline-CGPA scholarship, open coordinator inbox, confirm 5-field briefing renders	PASS, verified during finals build with the Demo Coordinator account
F5	Preview and edit decision letter with hallucination check	Click Preview & approve; modal shows letter, editable, regex check returns issues for any name/CGPA/year mismatch	PASS
F6	5-minute undo with mm:ss countdown	Approve a submission, countdown banner appears, click undo, status reverts	PASS, verified
F7	AI-suggest comment (3 pills)	Click Suggest comment, 3 AI-drafted pills appear, click one to insert	PASS

ID	Feature	Test case	Status
F8	Coordinator claim/release plus Mine filter	Click Claim, row tagged 'you', click Release, tag clears	PASS
F9	Bulk approve (excludes flagged)	Tick 3 unflagged plus 1 flagged; Bulk approve; flagged auto-excluded with reason summary	PASS
F10	Bulk request-info with editable note	Tick 2; Bulk request info; modal with editable note; confirm	PASS
F11	Real file upload via Supabase Storage	Drop PDF on file_upload step; file lands in application-files bucket; signed URL accessible only to owner and staff	PASS
F12	Auto-save per step (localStorage)	Type into a text step, reload, answer hydrates	PASS
F13	Realtime updates on applications, steps, letters, messages	Open student in tab A, coordinator in tab B; coordinator approves; A updates without refresh	PASS, verified after the migration that set REPLICA IDENTITY FULL
F14	Student-coordinator message thread (realtime)	Send message from coordinator; student sees it instantly	PASS
F15	Internal notes (staff-only)	Add internal note as coordinator; switch to student account; note not visible	PASS, RLS-enforced
F16	Step revise (student edits prior answer; AI re-plans)	Click Edit on Step 2, change answer, AI marks subsequent steps stale and re-emits	PASS
F17	Application withdraw	Withdraw a draft, status flips, coordinator inbox no longer surfaces it	PASS
F18	Notification bell with last 14 days plus unread count	Trigger 3 events, bell shows badge with count = 3	PASS
F19	Letter print (UM letterhead)	Click View letter, /letters/[id]/print loads, print to PDF works	PASS
F20	Admin SOP upload (paste, URL, PDF)	Upload all three modes, confirm chunks render in viewer	PASS, pdf-parse handles real-text PDFs;

ID	Feature	Test case	Status
			image-only rejected by design
F21	Admin letter template editor	Edit acceptance template, confirm new submissions use new wording; old letters unchanged	PASS
F22	Admin deadline editor	Set date plus label, student tile reflects it	PASS
F23	Admin analytics (KPIs plus per-procedure)	Visit /admin/analytics, KPIs render against seeded data	PASS
F24	Admin GLM trace viewer	Visit /admin/glm-traces, 100 most-recent calls listed; row expand shows input/output JSON	PASS
F25	Demo seed variety (9 sample applications)	Click any demo tile; /api/demo/reset fires; 9 apps in distinct states	PASS
F26	Mock mode plus demo banner plus auto-fallback	Unset ZAI_API_KEY; all calls return fixtures; banner appears	PASS
F27	Dual AI providers (Z.AI default; ILMU routable for briefing)	Set USE_ILMU_FOR_BRIEFING=true; submit application; trace shows ilmu-glm-5.1	PASS, provider tag visible in trace inputSummary
F28	Submit and decide endpoints degrade gracefully on AI failure	Force generateBriefing throw; /submit still flips status to submitted with fallback briefing	PASS, added fallback in this finals build
F29	AI reads uploaded PDFs (parseDocument wired)	Upload an EPF statement on Scholarship Step 3; response_data has extracted_fields keyed by extraction_schema; next step does not re-ask for income	PASS, added in finals build (commit 662bae8)
F30	Coordinator briefing cites specific SOP rules in flags	Submit a borderline application; flag message contains an SOP § reference	PASS, added in finals build (commit cff0032)
F31	Plan-mode confirmation on Request More Info	Type comment, click Preview info request, modal shows AI's proposed step, confirm or edit	PASS, added in finals build

ID	Feature	Test case	Status
			(commit f34cc1c)
F32	Coordinator coassist (revise letter, step, briefing)	In preview-letter modal, type 'make it softer', click Revise; letter updates inline; same pattern in step modal and briefing drawer	PASS, added in finals build (commit 9f14f6b)
F33	Always-on student AI chat with hybrid escalation	Type a question in the chat panel; AI answers; type 'I need a coordinator'; escalation pill appears; click; coordinator Triage tab surfaces it	PASS, added in finals build (commit 3ef77e1)
F34	/admin/cost token-spend dashboard	Visit /admin/cost; KPIs plus by-model plus by-endpoint plus 14-day trend render against live trace data	PASS, added in finals build (commit be77a4b)

4. Test Execution Evidence

The most recent end-to-end manual sweep was run on 2 May 2026 against the production-deployed finals build.

- **Student happy path (mock plus live mode):** Sign in via demo tile; demo data resets; open Draft scholarship application; walk Steps 1 to 6; reach the file_upload step with extraction_schema set; (live mode) confirm the AI's next step skips re-asking for income; submit; confirmation screen renders. PASS in both modes.
- **Coordinator decision flow:** Sign in via Coordinator demo tile; inbox lists 4 pending submissions sorted by AI urgency; open the highest-urgency row (block flag for declared income RM 2,000 vs EPF RM 5,800); briefing panel shows extracted facts, SOP-cited flags, AI recommendation, confidence score; click Preview & approve; modal renders AI-drafted letter; hallucination check runs and shows zero issues; click Send; status flips, 5-minute undo banner appears. PASS.
- **Coassist flow:** In preview-letter modal, type 'make this softer and add the appeal info paragraph'; click Revise; letter updates with softer wording plus appeal paragraph; brief explanation strip shows what changed; turn history populated. PASS in mock mode against the coassist_letter.json fixture.
- **Student chat plus escalation:** Type 'What is B40?'; AI answers from SOP, no escalation suggested. Type 'I don't have an EPF statement, can I use my parent's payslip'; AI suggests escalation; click pill; escalation_pending=true; coordinator Triage tab now shows the application even though it's a draft; coordinator opens it; escalation context card visible at top; reply in message thread; student sees reply in real time. PASS.
- **Demo PDF parsing (live mode):** Drop tests/fixtures/sample-documents/01_epf_statement_b40.pdf onto Scholarship Step 3 (file_upload with extraction_schema for income fields); click Continue; application_steps.response_data contains extracted_fields keyed by document_type,

account_holder, monthly_contribution, estimated_monthly_income_rm, statement_period, income_tier_inferred. PASS.

- **Mock-fallback resilience:** npm test:run; tests/glm-mock.test.ts passes; confirms every GLM and ILMU call site has a fixture and returns it cleanly when ZAI_API_KEY or ILMU_API_KEY is unset. PASS.

5. Defect Log

All defects opened during the finals build sub-phase. Closed items list the resolving commit; acknowledged items are deferred to post-finals with rationale.

ID	Defect	Severity	Status	Resolution
D1	next_step.md prompt didn't tell the AI to use parsed_attachments	Medium	Closed (commit 662bae8)	Updated prompt rule 2a to explicitly cover parsed_attachments plus low-confidence confirmation
D2	generateBriefing didn't receive sopChunks; flags weren't SOP-cited	Medium	Closed (commit cff0032)	Added sopChunks to GenerateBriefingInputSchema plus brief.md rule 6 plus submit/route.ts wiring
D3	Request More Info committed AI's step blindly without coordinator preview	Medium	Closed (commit f34cc1c)	Added /preview-step plus Preview info request modal plus step_override on /decide
D4	glm_reasoning_trace.endpoint CHECK constraint never widened; v2 endpoints' writes were silently bouncing	Medium	Closed (migration 0016)	Idempotent ALTER widens to v1 plus v2 plus coassist plus student_chat
D5	application_messages.author_role and author_id couldn't accept AI rows	Low	Closed (migration 0017)	Widened author_role check; nullable author_id; added kind column
D6	ApplicationData type missing escalation_pending field	Low	Closed (commit 3ef77e1)	Added to interface plus selected in /api/applications/[id] route
D7	tests/glm-mock.test.ts is the only automated test	Low (acknowledged)	Open (post-finals)	Integration tests plus end-to-end tests scheduled for Q3 2026 ramp; manual demo runs are the primary quality gate today
D8	7 moderate npm audit warnings in transitive dev deps	Low	Acknowledged (post-finals)	All gated behind --force which would downgrade Next 15 to Next 9. Wait for vitest patch

ID	Defect	Severity	Status	Resolution
				with non-breaking esbuild pin.

6. Risk Register

Top operational risks for the production deployment, with mitigation status. Severity is the post-mitigation residual; rated low, medium, or high relative to the demo and pilot operating envelope.

ID	Risk	Likelihood	Impact	Mitigation	Status
R1	AI hallucination in decision letter (wrong CGPA, name, programme)	High	High	Block or warn on regex hallucination check before send; coordinator-as-final-reviewer; never auto-send	Mitigated, checked at preview-letter time and after every coassist revise
R2	Z.AI provider rate-limit or downtime	Low	High	Mock-mode fallback returns recorded fixture; submit and decide endpoints write fallback briefing or letter rather than 502	Mitigated, tested by setting GLM_MOCK_MODE=true
R3	Schema migration breaks production	Low	High	All migrations idempotent (use IF NOT EXISTS); applied via SQL editor with manual review; daily Supabase snapshots	Mitigated, 0016 and 0017 applied cleanly to prod 2 May
R4	RLS policy hole exposes another user's data	Low	Critical	RLS on every PII table; service-role used only server-side; manual policy review on every migration	Monitored, no known holes
R5	Token cost grows unsustainably with usage	Medium	Medium	/admin/cost monitors live; route low-stakes calls to GLM-4.5-flash; cache common SOP retrievals	Monitored
R6	Demo collapses if AI	Low (with mitigation)	High	Mock-mode with auto-fallback fixtures;	Mitigated

ID	Risk	Likelihood	Impact	Mitigation	Status
	is unavailable during pitch			lavender banner makes mode obvious to judges	
R7	Hallucination check has a false negative (wrong fact passes through)	Medium	Medium	Coordinator IS the final reviewer; 5-min undo retracts mistakes; broader semantic check is on the post-finals roadmap	Acknowledged, coordinator gating plus undo are the primary controls
R8	Image-only PDFs fail extraction silently	Medium	Low	extractTextFromStorage returns null; AI just doesn't get extraction; the upload itself isn't blocked; OCR is on the roadmap	Mitigated
R9	AI proactively suggests too many escalations (fatigue)	Medium	Medium	Track escalation rate via /admin/glm-traces (student_chat endpoint); tune prompt if rate exceeds 30% of chats	Monitored, too early to assess
R10	Coassist over-revises a letter and removes a required clause	Low	Medium	Coordinator sees the full revised letter and clicks Send only after review; hallucination check re-runs after each revise	Mitigated

7. Architecture Trade-Offs

Decisions taken consciously for the hackathon time-box, and where they would be revisited.

- **Manual demo over comprehensive automated tests.** With under 22 hours to ship 8 commits plus 4 deliverable PDFs, integration test authoring would have crowded out the parseDocument wiring and the schema migrations. Quality is held by static typing plus manual sweeps plus resilience fallbacks. Post-finals: build the test pyramid out properly.
- **All-SOP-chunks retrieval over pgvector semantic search.** Semantic search adds embedding cost plus a kNN call per step. With 5 procedures times about 10 chunks each, the full SOP fits in context. Activate pgvector when the catalogue exceeds 30 procedures.
- **Inline parseDocument over an attachments table.** Storing extracted_fields directly on the step's response_data avoids a schema migration and simplifies buildHistory. Multiple files per step are out of scope today (max_files default 1). The schema field parsed_attachments is already an array, so a future attachments table can swap in transparently.

- **Single unified application_messages table over a separate ai_chat_messages table.** The chat thread is one continuous conversation between student, AI, and coordinator. Splitting tables would have forced UI joins and double-realtime subscriptions. Widening author_role to include 'ai' was a one-migration change.
- **Hybrid escalation (status unchanged) over auto-submit.** Auto-submit conflates 'I have a question' with 'this is finished'. The hybrid model preserves the formal flow while making coordinators aware. Trade-off: requires a Triage inbox tab plus a per-row badge.
- **Coassist as a chat (per-modal) over an editor (per-document).** Chat history is ephemeral by design; once the coordinator clicks Send, the modal closes and the iteration history goes with it. Persisting the chat would create an audit-trail expectation we don't yet have a policy for. The committed letter or step is the source of truth.

8. Resilience Test Cases (Deep Dive)

Each fallback path was deliberately exercised during the finals build to confirm graceful degradation.

- Set ZAI_API_KEY to an invalid value, then submit a scholarship application. Observe: every GLM call falls back to its named mock fixture; trace rows are written with model='glm-4.6-fallback'; demo banner does NOT appear (live mode is preserved). Application advances normally.
- Set ZAI_API_KEY to an invalid value AND let submit fail on briefing generation. Observe: /submit returns success with briefing_fallback=true; application_briefings row written with recommendation='review' and a single warn flag asking the coordinator to regenerate; status flipped to submitted. Application appears in inbox.
- Mock GLM returns malformed JSON for nextStep. Observe: nextStep retries once with a corrective prompt; if still malformed, the call throws (the engine catches and surfaces a 500). This is the only path that does not have a quiet fallback, by design, since it indicates the model is fundamentally not following the schema and the operator should investigate.
- Image-only PDF uploaded on a file_upload step. Observe: extractTextFromStorage returns null (text length below threshold); /respond proceeds with the original responseData; no error to the student; the AI just doesn't get the extraction for that step.
- Coassist call fails on a letter revision. Observe: the modal shows an error banner; the coordinator's edits to the letter text are NOT lost; they can retry or skip the revision and proceed to Send.
- Student chat fails (provider unreachable). Observe: a fallback AI reply is inserted that itself suggests escalation, so the student has an action; the message is persisted normally so the coordinator sees it.
- Trace insert fails (DB connection error). Observe: writeTrace catches and logs; the user-facing call result is unaffected. Audit data for that one call is lost; user is not blocked.

9. Quality Gates

Policy gates that must pass before code is merged to main.

- npm run typecheck must be clean (TypeScript strict mode).
- npm run lint must be clean (Next.js lint, no errors).
- npm run test:run must be green (vitest, smoke test passes).
- Manual happy-path sweep on the affected surface or surfaces before merging non-trivial changes.

- Schema migrations must be idempotent (use IF NOT EXISTS plus conditional drops); reviewed by the technical lead; applied to staging before production.
- Prompt edits will change the prompt_hash in subsequent traces (this is desired; it's the audit signal that explains shifts in AI output behaviour).

10. Future Test Plan (Post-Finals)

- **Q3 2026, integration tests:** vitest plus supertest covering each /api/ route's happy path and error paths against a staging Supabase. Target: 70% line coverage of /lib/ plus /app/api/.
- **Q3 2026, end-to-end tests:** Playwright covering the three role surfaces' standard-case flows against a deployed preview environment. Run on every PR via Vercel Checks.
- **Q4 2026, accessibility tests:** axe-core integration via Playwright; WCAG 2.1 AA target across all student-facing surfaces.
- **Q4 2026, load tests:** k6 scenarios for the coordinator inbox under 1,000 plus active applications and the student smart-application under realistic burst patterns.
- **Q1 2027, chaos tests:** inject Z.AI 5xx, Supabase connection drops, and storage failures on staging; verify all fallbacks engage.

11. Terms & Definitions

Recurring jargon used throughout this document, defined in plain language.

Term	Plain-language definition
AI hallucination	When the AI states something untrue or invents a fact (for example, the wrong CGPA in a letter). UniGuide's hallucination check is a regex layer that catches these mismatches before any letter is sent.
Coassist	UniGuide's natural-language Revise channel for coordinators: type 'make it softer' or 'add the appeal info' and the AI revises the letter, the proposed step, or the briefing reasoning inline.
Defect log	A list of bugs found during testing, each with severity, status (open / closed / acknowledged), and the resolving commit (when closed).
End-to-end test	An automated test that exercises the full system from the user's perspective (clicking buttons in a real browser), as opposed to testing one function or one endpoint in isolation.
Five-minute undo	After a coordinator approves or rejects an application, they have a five-minute window with a live countdown during which they can take the decision back before it commits permanently.
GLM	Z.AI's family of large language models (GLM-4.6 and GLM-4.5-flash). The mandatory model for UMHackathon 2026.

Term	Plain-language definition
Happy path / standard case	The straightforward case where nothing unusual happens (e.g. a student with high CGPA and B40 income tier applying for the standard scholarship). Contrast with edge cases needing human discretion.
ILMU	Malaysia's sovereign large language model, developed by YTL AI Labs and Universiti Malaya. Routable as the briefing provider in UniGuide for Bahasa-heavy content.
Integration test	An automated test that exercises one API route end-to-end against a real database, but without driving a browser. Sits between unit tests and end-to-end tests.
kNN search	k-Nearest-Neighbours search; finding the k items most similar to a query item by some distance metric. Used in semantic SOP retrieval (planned, not yet active).
Mock fallback	Recorded sample AI responses (in tests/fixtures/glm/) that the application returns when the real AI provider is unavailable, so the demo never collapses.
Pgvector	A Postgres extension that stores and searches vector embeddings, used for semantic similarity search. Installed but not yet active in UniGuide; ready for the next scale step.
Prompt hash	A short fingerprint of the AI prompt used for a particular call. Recorded on every trace so reproducibility across prompt revisions is auditable.
RLS	Row-Level Security; a Postgres feature that restricts which rows each user can read or write. UniGuide enforces RLS on every table holding personal data.
Smoke test	A quick automated test that exercises the most critical paths to confirm the system is broadly working. UniGuide's smoke test exercises the AI mock-fallback paths.
SOP	Standard Operating Procedure; the official UM document describing how a procedure (scholarship, FYP, deferment, etc.) is meant to run. UniGuide's AI reads the SOP at every step.
Test pyramid	A common testing-coverage shape with many fast unit tests at the base, fewer integration tests in the middle, and a small number of slow end-to-end tests at the top.
Trace row	A row in the glm_reasoning_trace table recording one AI call: endpoint, model, prompt hash, input summary, output, latency, tokens, confidence, retry count.

Term	Plain-language definition
Triage	In coordinator UX: the inbox tab that surfaces applications a student has flagged for human help, regardless of whether the application has been formally submitted yet.

12. Conclusion

UniGuide at finals submission is a deliberately scoped MVP with conscious trade-offs in test depth offset by static-typing rigour and resilience-first engineering. Every PRD feature plus six new features added in the finals build are traced to a test case and a current PASS. The post-finals quality plan ramps automated coverage to a full pyramid without rewriting any of today's surface code; the seams are already in place.

End of Refined QATD